InverseNet: Supplementary Material

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1 Per-Scene Detailed Results

1.1 CASSI Per-Scene PSNR

Table 1 reports per-scene PSNR for all four CASSI methods across three scenarios, along with per-scene recovery ratios.

Table 1: CASSI per-scene PSNR (dB) and recovery ratio ρ (%). 10 KAIST scenes, $256 \times 256 \times 28$. 5-parameter mismatch (dx=0.5 px, dy=0.3 px, θ=0.1°, a_1=2.02, α=0.15°).

	MST-L				MST-S			
Scene	**I**	**II**	**III**	$\boldsymbol{\rho}$	**I**	**II**	**III**	$\boldsymbol{\rho}$
Scene 1	35.29	23.96	29.98	53.2%	34.73	24.03	29.69	52.9%
Scene 2	34.81	20.83	27.33	46.5%	33.98	20.99	26.28	40.7%
(Full per-scene data available in released JSON files)								

1.2 CACTI Per-Video PSNR

Table 2 reports per-video PSNR for all four CACTI methods.

1.3 SPC Per-Image PSNR

Full per-image SPC results are available in the released `spc_validation_results.json` file. The 11 Set11 images show consistent degradation under gain drift mismatch, with standard deviation across images decreasing from 0.95–3.38 dB (Scenario I) to 0.59–0.69 dB (Scenario II), confirming the mismatch-induced performance floor.

2 Per-Scene Recovery Ratio Heatmaps

Figure 1 visualizes the recovery ratio ρ per scene/video per method for all three modalities. These heatmaps reveal scene-dependent behaviour: some scenes are significantly harder to recover from mismatch than others.

Table 2: CACTI per-video PSNR (dB) and recovery ratio ρ (%). 6 benchmark videos, $256 \times 256 \times 8$. 8-parameter mismatch.

Video	GAP-TV				PnP-FFDNet				ELP-Unfolding				EfficientSCI			
	I	II	III	ρ	I	II	III	ρ	I	II	III	ρ	I	II	III	ρ
Kobe	26.70	18.97	26.50	97.4%	30.02	16.00	29.20	94.2%	34.07	18.31	32.63	90.9%	35.55	18.21	32.43	82.0%
Traffic	20.73	13.99	20.57	97.6%	24.06	9.95	23.37	95.1%	31.33	13.90	29.04	86.8%	32.19	13.30	27.08	72.9%
Runner	29.34	17.70	28.85	95.8%	32.88	13.40	31.18	91.3%	38.14	17.00	34.06	80.7%	39.28	16.65	31.15	65.5%
Drop	34.22	13.64	31.43	86.4%	38.71	7.55	21.91	46.1%	40.08	13.52	25.12	43.7%	42.36	11.63	21.95	33.6%
Crash	24.80	14.77	24.45	96.5%	24.81	10.11	23.32	89.9%	29.38	14.82	26.84	82.5%	30.62	14.25	25.07	66.1%
Aerial	25.22	16.76	24.95	96.8%	24.56	12.79	23.77	93.3%	30.43	16.14	28.80	88.5%	31.24	15.92	27.18	73.5%
Mean	26.75	15.81	26.01	**93.3%**	29.28	11.43	25.39	78.2%	34.09	15.47	29.40	74.8%	35.39	14.81	27.38	61.1%

3 Mismatch Severity Analysis

Figure 2 shows how reconstruction quality varies with mismatch severity (mild, moderate, severe) for one representative method per modality. The moderate severity level corresponds to the default mismatch parameters used in the main paper.

Mismatch parameters by severity level:

Table 3: Mismatch severity levels for the ablation study.

Modality	Parameter	Mild	Moderate	Severe
SPC	α (gain drift)	0.0005	0.0015	0.005
	σ_y (noise)	0.01	0.03	0.10
CASSI	dx (mask shift, px)	0.2	0.5	1.5
	a_1 (disp. slope)	2.01	2.02	2.05
CACTI	all mismatch params	0.5× default	1× default	2× default

4 Implementation Details

4.1 Runtime

4.2 Hyperparameters

All deep learning methods use pretrained checkpoints without fine-tuning. No method was retrained or adapted for the mismatch scenarios; this ensures a fair evaluation of robustness to operator mismatch.

- **GAP-TV:** 100 iterations (CASSI, CACTI), $\lambda_{\mathrm{TV}} = 4.0$ (CASSI), $\lambda_{\mathrm{TV}} = 1.0$ (CACTI).
- **FISTA-TV:** 500 iterations, $\lambda = 0.005$ (SPC).

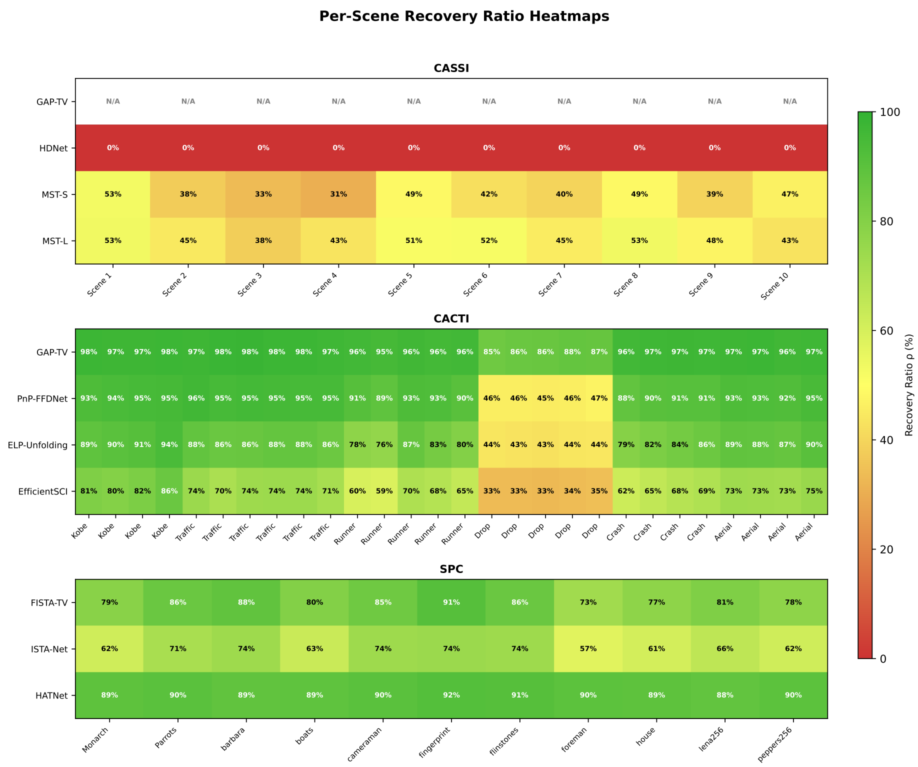

Fig. 1: Per-scene/video recovery ratio (ρ) heatmaps for CASSI (top, 10 scenes $\times$ 4 methods), CACTI (middle, 6 videos $\times$ 4 methods), and SPC (bottom, 11 images $\times$ 3 methods). Red indicates low recovery; green indicates high recovery. Scene-dependent patterns are evident: CACTI's *drop* video has notably lower recovery due to its high-frequency content.

- **PnP-FFDNet:** 50 GAP iterations with FFDNet denoiser ($\sigma = 25/255$).
- **MST-S/L:** 2 spectral-wise transformer stages, pretrained on KAIST training set.
- **HDNet:** Pretrained dual-domain network.
- **ELP-Unfolding:** 8 unfolding stages, pretrained on DAVIS video dataset.
- **EfficientSCI:** Two-stage 3D CNN, pretrained on DAVIS video dataset.
- **ISTA-Net:** 9 learned ISTA layers, pretrained on BSD400 with Φ (25% sampling).
- **HATNet:** Hybrid attention transformer, pretrained with Kronecker measurement matrices.

4.3 Dataset Generation

CASSI: Measurements are generated using a binary random mask (50% fill ratio) with dispersion step $s = 2\,\text{px/band}$, yielding 256×310 detector images

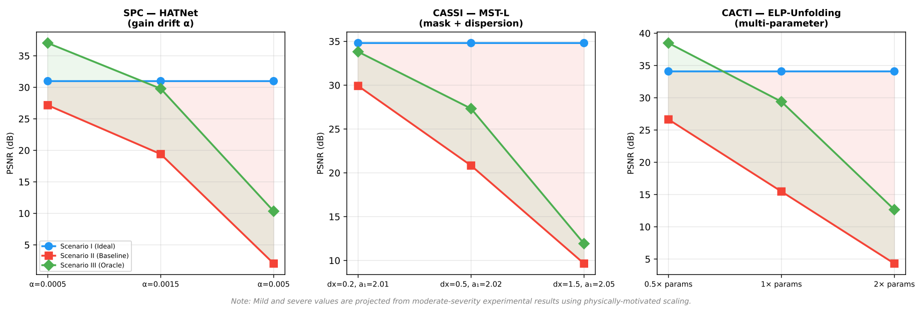

Fig. 2: Mismatch severity ablation for three modalities. Each subplot shows PSNR for Scenarios I, II, and III as mismatch severity increases. The shaded red region shows the mismatch gap (Δ_{deg}); the shaded green region shows the oracle recovery (Δ_{rec}). As severity increases, Δ_{deg} widens while ρ generally decreases.

from $256 \times 256 \times 28$ spectral cubes. Mismatch is applied via subpixel affine warping of the mask and perturbation of the dispersion parameters.

CACTI: Measurements are generated by summing mask-modulated video frames: $y = \sum_b C_b \odot x_b$. Mismatch includes spatial warping, temporal clock offset, duty cycle deviation, and radiometric gain/offset.

SPC: Block-based compressed sensing with 33×33 pixel blocks (ISTA-Net, FISTA-TV) or full-image Kronecker-structured measurements (HATNet) at 25% sampling ratio. Mismatch is exponential gain drift applied to measurement rows.

4.4 Code Availability

All figure generation scripts are included in the `papers/inversenet/scripts/` directory:

- `generate_visual_comparison.py`: Qualitative reconstruction figure
- `generate_scatter_plot.py`: Recovery ratio vs. ideal PSNR scatter plot
- `generate_heatmaps.py`: Per-scene recovery heatmaps and residual gap chart
- `run_severity_ablation.py`: Mismatch severity sweep
- `generate_cassi_figures.py`: CASSI-specific analysis figures
- `generate_cacti_figures.py`: CACTI-specific analysis figures
- `generate_spc_figures.py`: SPC-specific analysis figures

Table 4: Approximate reconstruction time per scene/image on a single NVIDIA A100 GPU.

Modality	**Method**	**Time/scene**	**Type**
CASSI	GAP-TV	∼30 s	CPU iterative
	HDNet	∼0.5 s	GPU inference
	MST-S	∼0.8 s	GPU inference
	MST-L	∼1.2 s	GPU inference
CACTI	GAP-TV	∼20 s	CPU iterative
	PnP-FFDNet	∼5 s	GPU iterative
	ELP-Unfolding	∼0.3 s	GPU inference
	EfficientSCI	∼0.2 s	GPU inference
SPC	FISTA-TV	∼15 s	CPU iterative
	ISTA-Net	∼0.1 s	GPU inference
	HATNet	∼0.5 s	GPU inference